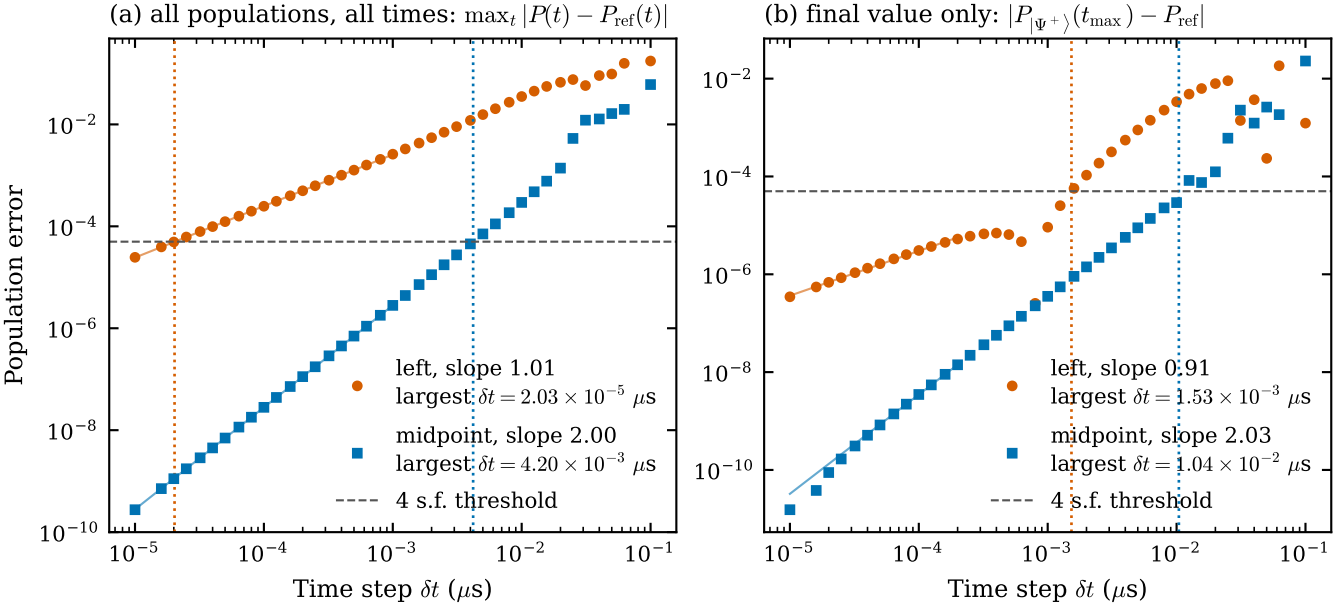
(a) all populations, all times: $\max_t |P(t) - P_{\mathrm{ref}}(t)|$
(b) final value only: $|P_{|\Psi^+\rangle}(t_{\max}) - P_{\mathrm{ref}}|$
Population error
Time step $\delta t$ ($\mu$s)
left, slope 1.01
largest $\delta t = 2.03 \times 10^{-5}$ $\mu$s
midpoint, slope 2.00
largest $\delta t = 4.20 \times 10^{-3}$ $\mu$s
4 s.f. threshold
left, slope 0.91
largest $\delta t = 1.53 \times 10^{-3}$ $\mu$s
midpoint, slope 2.03
largest $\delta t = 1.04 \times 10^{-2}$ $\mu$s
4 s.f. threshold